TETHER: Tethered Causal Temporal Refinement for Frozen Surgical Stereo

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***Abstract*— Dense stereo is attractive for robotic surgery because it provides metric geometry without active illumination, yet modern stereo estimators are typically applied independently to each frame and can therefore fluctuate under specularities, weak texture, tissue deformation, tool occlusions, and camera motion. We ask whether temporal information can improve frozen stereo predictions without modifying the stereo estimator, accessing its internal representation, or using future frames.**

We present TETHER, a causal plug-and-play temporal refiner for frozen stereo. At each frame, preceding disparity is motion-aligned to the current view using frozen SEA-RAFT, and a shared 142-channel external evidence representation feeds a 177,338-parameter CODD-inspired reset-and-fusion head that predicts a bounded interpolation between the current stereo estimate and that aligned memory. Corrected recurrence is explicitly bounded and periodically re-anchored to independently predicted raw geometry.

A controlled development-set closure shows that the learned policy is not explained by temporal smoothing: on SCARED-C D2, canonical $H = 4$ reduces mean EPE by 4.05%, compared with only 0.59% for the best matched fixed/EMA blend, while direct flow-confidence weighting and ungated memory propagation degrade accuracy. The raw-versus-memory oracle exposes a 12.30% improvement ceiling, of which TETHER recovers 37.2%. Continuous corrected-state recurrence increases EPE by 62.69%, supporting bounded memory.

One frozen $H = 4$ checkpoint subsequently improves all five evaluated stereo estimators on held-out SCARED-C D7, including CREStereo and Fast-FoundationStereo, which were excluded from temporal-head training. Mean backbone EPE decreases from 0.4921 to 0.4696, while Bad1 and disparity RMSE also decrease for every backbone. Zero-shot evaluation on five DRENDS recordings ($20{,}929$ frames) with three estimators, two of them unseen in training, improves disparity EPE, Bad3 and metric depth in all 60 recording–backbone–metric cells, completing the transfer matrix across both the backbone and the domain shift. We further show that confining the fused output to the support the module already declares is load-bearing rather than cosmetic: without it, zero-padded out-of-bounds memory turns rare pixels into invalid predictions and produces the metric-depth tail previously attributed to the external domain.

We additionally introduce a refinement-centric evaluation framework measuring spatial geometry, multi-horizon temporal fidelity, introduced and recovered error, frame tails and selective risk under fixed, prediction-independent support.

I. Introduction

Metric three-dimensional perception underpins camera navigation, tool–tissue distance estimation, augmented reality and deformation analysis in robot-assisted surgery, and stereo endoscopy is attractive because depth is recovered passively from images the platform already provides.

It is also hard. Endoscopic scenes carry weak texture, non-Lambertian tissue, specular highlights, blood, smoke, deformable anatomy, rapid viewpoint change and frequent tool occlusion [18], [19]. Modern estimators [11]–[15] give increasingly strong frame-wise disparity, but nothing in a frame-wise model requires its predictions to cohere over time. Video stereo methods address this by building temporal reasoning into the model [1]–[4], at the price of assuming a temporal architecture, model-specific representations, future observations, or co-design with the estimator.

We study a different deployment setting: *can one causal temporal module be trained once and attached unchanged to heterogeneous frozen stereo estimators?* This matters because backbone and temporal logic can then evolve independently: a new estimator replaces the previous one without the temporal module needing its cost volume, hidden state, confidence head or training pipeline.

Our starting point is the reset-and-fusion principle of CODD [1], which we do not claim as novel. We keep its inductive bias — decide whether motion-aligned history should influence the current estimate, and constrain the correction to a convex interpolation between the two — and redesign it for a black-box interface: the integrated motion pathway becomes frozen SEA-RAFT, backbone-specific representations become a shared external evidence space, and the same head is evaluated unchanged across architectures.

The structure is deliberately constrained. The current estimate stays an anchor, temporal memory is geometrically aligned before use, the head controls how much to intervene but cannot leave the interval spanned by the two candidates, and corrected recurrence is periodically reset to raw geometry.

These restrictions raise a question: does the learned head add anything that simple temporal averaging would not? We answer it experimentally under a frozen development protocol. Matched fixed fusion, EMA, warped-memory propagation, and direct forward–backward confidence weighting recover only a small fraction of TETHER's gain. In contrast, a ground-truth oracle shows that the aligned memory contains substantially more useful information than the current policy exploits. This indicates that the core problem is not obtaining temporal evidence, but identifying *where and how much* of it should be used.

A second question is how such intervention should be evaluated. Smoothness is not correctness — a stale prediction can be smooth and geometrically wrong — and lower mean EPE does not reveal whether refinement introduces new bad pixels or rare catastrophic frames. We therefore evaluate refinement relative to the frozen upstream prediction and explicitly separate geometry, temporal fidelity, benefit, harm, and tail behavior.

Our contributions are three.

(i) We formulate surgical temporal stereo refinement as a *causal plug-and-play adaptation problem* over frozen, heterogeneous estimators, needing neither future frames nor backbone internals, and instantiate it as TETHER: a 177,338-parameter reset-and-fusion head over causal SEA-RAFT alignment, a shared 142-channel external evidence space, support-confined convex fusion and bounded corrected-state recurrence.

(ii) We show through a matched-baseline closure that the gain is not temporal smoothing — 4.05% against 0.59% for the best fixed blend under an identical state machine — and that one frozen checkpoint transfers along both axes: to estimators excluded from training on held-out D7, and to an unseen domain, the first measurement of the two shifts combined.

(iii) We introduce a refinement-centric evaluation protocol under fixed, prediction-independent support that measures introduced as well as recovered error, and show what it catches that geometry and smoothness scores do not — applying it to our own claims first: degenerate history-replay controls win the no-reference consistency metric, and the head's own weight inverts its relation to harm across splits.

II. Related Work

a) Deep stereo matching.: Stereo has moved from cost-volume regression to iterative recurrent estimation [11], [12] and broad-domain zero-shot models [13]–[15]. Our work is orthogonal: we neither modify the estimator nor assume an architecture.

b) Temporal stereo and depth.: CODD introduced the reset/fusion mechanism we build on [1]; DynamicStereo [2], PPMStereo [4] and TC-Stereo [6] follow, all co-designed with the estimator, so a new backbone means a new temporal model. Match Stereo Videos exposes BiDAStabilizer as a plug-in on a frozen estimator [3], but propagates bidirectionally. A parallel line enforces consistency on monocular video depth by diffusion [24] or stabiliser networks [16]: a different output space, none causal at clip level. A further line adapts the stereo weights online [25], the opposite of our constraint.

c) What can actually be compared.: We are aware of no runnable published method in this exact setting — causal, black-box, frozen weights. The closest, StereoDiffusion [7], ships no implementation. Of the rest, [8] and [9] publish code whose weights are no longer retrievable, and TC-Stereo warps its temporal state by a rigid reprojection needing per-frame pose, intrinsics and baseline — privileged inputs our interface does not assume, though SCARED-C supplies them. We compare against the one runnable published method under our own constraints, BiDAStabilizer (Sec. V-G), which sees the future while we do not.

d) Evaluating temporal refinement.: Stereo benchmarks report frame-wise geometry and video methods add consistency measures, but neither characterises an intervention: a prediction can grow smoother without growing more correct, and a lower mean error can hide newly created failures. Recovered and introduced error must therefore both be measured, which connects to selective prediction [17] and to the uncertainty literature asking when a prediction should be trusted [26]. Our contribution is not a new metric but a common protocol under fixed, prediction-independent support.

e) Surgical stereo geometry.: SCARED [18], SERV-CT [19], D4D [21] and DRENDS [22] supply the endoscopic stereo, CT-derived and structured-light geometry used here; Sec. IV states what each can and cannot support. Downstream reconstruction and tracking pipelines consume exactly the per-frame geometry we refine [20], [27], which is why we treat refinement as an intervention on that geometry.

III. Method

A. External Causal Interface

A frozen estimator S produces $d_t^{\text{raw}}, M_t^{\text{raw}} = S(I_t^L, I_t^R)$, positive-left disparity and its validity mask. The temporal module never reads the internal representation of S; it consumes only externally observable quantities — RGB, disparity, validity, optical flow and evidence derived from them. At time t the cache holds the preceding left image, the preceding raw disparity and validity, the previous fused state and its age; nothing from $t+1$ onwards exists. The map to learn is therefore $d_t^{\text{fused}} = T_\theta(d_t^{\text{raw}}, m_{t-1}, I_t^L, I_t^R, I_{t-1}^L)$.

B. Causal Motion Alignment

Frozen SEA-RAFT [5], [23] estimates both directions between the two already observed left images, $F_{t\to t-1} = F(I_t^L, I_{t-1}^L)$ and $F_{t-1\to t} = F(I_{t-1}^L, I_t^L)$; the second is used only to measure reliability. Bidirectional *flow estimation* between two observed frames is not bidirectional temporal inference.

Previous geometry is pulled onto the current grid, $\widetilde{m}_t(x) = m_{t-1}(x + F_{t\to t-1}(x))$, by bilinear sampling with zero padding and `align_corners=True`. Warp support $\mathcal{S}_t$ records whether the source coordinate lies inside the preceding image, and historical validity is sampled on the same grid. Pulling the reverse flow the same way, $\widetilde{F}_{t-1\to t} = \mathcal{W}(F_{t-1\to t}, F_{t\to t-1})$, gives the forward–backward residual $e_t^{FB} = \|F_{t\to t-1} + \widetilde{F}_{t-1\to t}\|_2$ and the reliability cue

$$C_t^{FB} = S_t \exp\left(-\frac{e_t^{FB}}{0.5 + 0.01\left(\|F_{t\to t-1}\|_2 + \|\widetilde{F}_{t-1\to t}\|_2\right)}\right). \tag{1}$$

The zero padding in this warp is not a detail: it is what makes $\widetilde{m}_t$ exactly zero outside the source image, and Section V-J shows what happens when that value is allowed into the output.

C. Shared External Evidence

All stereo estimators are mapped into one common evidence representation:

$$C_t = \Phi\left(d_t^{\text{raw}}, \widetilde{m}_t, I_t^L, I_t^R, I_{t-1}^L, F_{t\to t-1}, C_t^{FB}, S_t, \widetilde{M}_t\right). \tag{2}$$

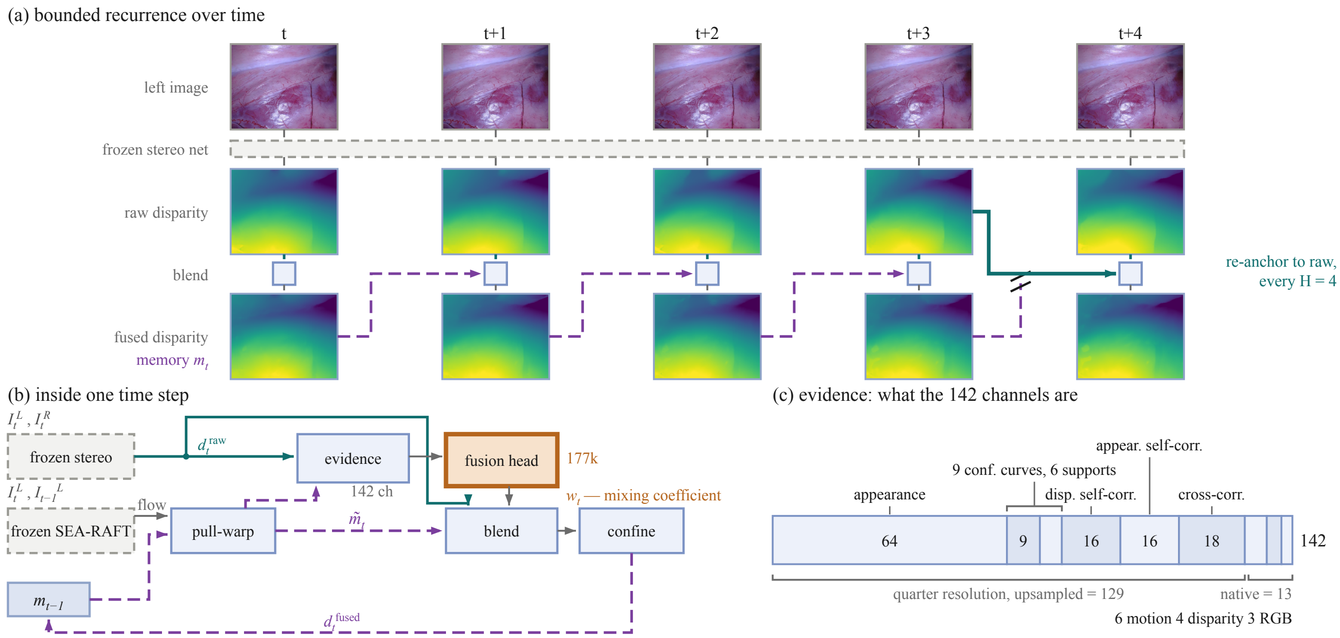

Fig. 1. TETHER at three scales. **(a)** Bounded recurrence. Temporal memory carries the *fused* disparity forward, and every $H{=}4$ frames the chain is cut and re-tied to raw geometry the frozen network produced independently (teal). The five frames are consecutive and the two disparity rows share one colour scale: they differ by 0.108 px on average, because the intervention is sub-pixel by design. **(b)** One time step. Dashed grey is frozen; the single orange block is the only trained component, and it emits one scalar w_t rather than geometry. **(c)** The evidence space: none of the 142 channels carries ground truth, backbone identity or a future frame, which is what lets one head serve heterogeneous estimators.

The head consumes a single 142-channel tensor built at two resolutions, itemised in Fig. 1(c): 129 quarter-resolution channels of matching and frozen appearance evidence, including 64 frozen ResNet-18 layer-1 channels [10], upsampled and concatenated with 13 cache-grid channels of disparity, motion and RGB.

Normalisation is fixed before training: disparities clipped to $[0, 64]$ and scaled by 64, signed residuals clipped to $[-16, 16]$ and scaled by 16, flow components on $[-32, 32]$ and magnitude on $[0, 32]$, and ImageNet-normalised RGB for the frozen features. No ground truth, backbone identity or future-frame signal enters this space, which is what lets one head serve heterogeneous estimators.

D. Reset-and-Fusion Decision Head

The decision structure is inspired by CODD [1]. We do not claim reset/fusion or convex interpolation as novel in isolation.

The cache-grid pathway maps the 142-channel input to 24 channels through a $3{\times}3$ convolution, GroupNorm, SiLU, and a residual block. A coarse pathway expands to 48 channels at one-quarter resolution, followed by three 48-channel residual blocks.

A full-resolution reset branch predicts $r_t = \sigma(z_t^r)$, while the coarse fusion branch predicts $f_t = \uparrow_4 [\sigma(z_t^f)]$. Running the context branch coarsely is a compute choice, not a requirement: at identical parameter count it costs about $16\times$ less than the full-resolution alternative, which Sec. V-I shows does not perform worse.

The effective temporal weight is $w_t = r_t f_t$.

Temporal evidence is only defined where the aligned memory is itself usable. We therefore define the per-pixel *support* as the conjunction of the current raw validity, the aligned temporal validity, and the warp support,

$$S_t = M_t^{\text{raw}} \wedge \widetilde{M}_t \wedge \mathcal{S}_t, \tag{3}$$

— the warp support $\mathcal{S}_t$ of the previous paragraph, the raw validity and the sampled historical validity — and confine the intervention to it. Final disparity is

$$\boxed{d_t^{\text{fused}} = \begin{cases} (1 - w_t) d_t^{\text{raw}} + w_t \tilde{m}_t, & S_t = 1, \\ d_t^{\text{raw}}, & S_t = 0. \end{cases}} \tag{4}$$

Because $w_t \in [0, 1]$, on the support

$$\min(d_t^{\text{raw}}, \tilde{m}_t) \le d_t^{\text{fused}} \le \max(d_t^{\text{raw}}, \tilde{m}_t), \tag{5}$$

and off the support the module is the exact identity on raw geometry.

The refiner therefore cannot synthesize arbitrary disparity outside the interval spanned by its two candidates.

Applying (3) to the output — rather than only computing it — is a load-bearing part of the contract, not a formality. The causal pull-warp uses zero padding, so an out-of-bounds sample returns $\tilde{m}_t = 0$ *exactly*. An unmasked convex blend at such a pixel can therefore drive d_t^{fused} to zero, producing an invalid disparity where the frozen stereo estimator produced a valid one, and the bounded recurrence of Sec. III-E then carries that zero forward until the next raw re-anchor. Section V-J quantifies the effect: it is invisible on SCARED-C and dominates the external-domain metric-depth tail.

The exact trainable parameter count is 177,338, and masking changes no parameter, no threshold, and no training procedure.

Reset output bias is initialized to -2, while fusion is initialized with zero bias, producing conservative initial temporal intervention.

E. Bounded Corrected-State Recurrence

After refinement, $m_t = d_t^{\text{fused}}$.

This creates the feedback path

$$d_{t-1}^{\text{fused}} \to \widetilde{m}_t \to d_t^{\text{fused}} \to \widetilde{m}_{t+1}. \tag{6}$$

Continuous propagation can therefore transform previous alignment and fusion errors into future evidence.

We explicitly bound corrected-state lifetime. On a sequence boundary the first output is raw. After a re-anchor, previous raw disparity is used as state. Otherwise, the previous fused output is propagated.

The canonical policy permits four corrected-state uses before a raw re-anchor. Thus H denotes state lifetime rather than a temporal input window.

The method remains one-step causal: $\{I_{t-1}, I_t, m_{t-1}\} \to d_t^{\text{fused}}$.

F. Training Objective

For target disparity g_t let $e_t^{\text{raw}} = |d_t^{\text{raw}} - g_t|$, $e_t^{\text{mem}} = |\widetilde{m}_t - g_t|$ and $\Delta_t = e_t^{\text{mem}} - e_t^{\text{raw}}$, so $\Delta_t < 0$ marks pixels where aligned memory is the better candidate. Training support M is the intersection of GT coverage, current validity, aligned temporal validity and warp support.

The objective is $\mathcal{L} = \mathcal{L}_{disp} + \mathcal{L}_{reset} + \mathcal{L}_{fusion}$. The reconstruction term is $\mathcal{L}_{disp} = \mu_M[\text{Huber}_{\delta=1}(d_t^{\text{fused}} - g_t)]$. The two decision terms supervise the branches directly against Δ_t at native margins $\tau_r^n = 5\,\text{px}$ and $\tau_f^n = 1\,\text{px}$ (cache scale 0.703125 and 0.140625): writing $R^{\pm}$ and $F^{\pm}$ for the sets where Δ_t exceeds $\pm\tau_r^c$ and $\pm\tau_f^c$, and F^0 for $|\Delta_t| \le \tau_f^c$,

$$\mathcal{L}_{reset} = \mu_{R^+}[r_t] + \mu_{R^-}[1 - r_t], \tag{7}$$

$$\mathcal{L}_{fusion} = \mu_{F^+}[f_t] + \mu_{F^-}[1 - f_t] + 0.2\,\mu_{F^0}[|f_t - 0.5|]\,. \tag{8}$$

No explicit temporal-consistency objective is used: temporal behaviour is a consequence of the decision, never a training target.

IV. Experimental Protocol

A. Canonical Training

The canonical temporal refiner is trained on temporally curated SCARED-C data derived from SCARED [18].

Training uses frozen predictions from S^{2}M^2-S [14], RAFT-Stereo [11] and Stereo Anywhere [13]; CREStereo [12] and Fast-FoundationStereo [15] are excluded from temporal-head training, and no backbone identity is ever given to TETHER.

The canonical configuration trains on D1, D3 and D6 with D2 for development and D7 held out, over 8,616 frames giving 8,606 causal pairs from three seen backbones, using non-overlapping $H = 4$ clips, batch size 4, 12 epochs of AdamW at 2×10^{-4} with weight decay 10^{-4} and gradient clipping 1.0. The 177,338 trainable parameters are selected on development fused EPE.

B. External Evaluation

a) DRENDS.: DRENDS contains dynamic robotic-endoscopy sequences with stereo imagery, calibration, and metric geometry [22]. We report five curated recordings — `Vid10_Liver_Med`, `Vid11_Liver_High`, `Vid12_Pancreas_Ext`, `Vid13_Pancreas_Med` and `Vid14_Pancreas_High` — evaluated with three stereo estimators for 20,929 frames in total. The zero-shot claim applies only to the SCARED-trained canonical checkpoint: no DRENDS data enters training, checkpoint selection or threshold calibration.

Disparity reference is derived from metric ToF geometry through stereo calibration. It is therefore useful metric evidence but not independent stereo ground truth, and the ToF reference itself is temporally filtered.

b) D4D.: D4D contains deformable surgical stereo sequences and structured-light observations [21]. Because the current geometric scale contract remains unresolved, we use D4D only for no-reference temporal diagnostics.

c) SERV-CT.: SERV-CT provides calibrated stereo pairs and CT-derived reference geometry [19]. The evaluated pairs are non-consecutive, so temporal refinement is marked not applicable.

C. Unified Temporal-Refinement Evaluation

A central requirement is that evaluation support cannot depend on the prediction: $M_{\text{eval}} = M_{\text{GT}} \cap M_{\text{protocol}}$.

Invalid predictions on valid support are penalized rather than silently removed.

The framework evaluates five complementary dimensions.

a) Spatial geometry.: For disparity we report EPE/MAE, median, RMSE, P90/P95/P99, maximum, invalid rate, and Bad-1/3/5. For metric depth we additionally report Bad-2/5/10 mm, AbsRel, SqRel, and $\delta_{1,2,3}$.

b) Temporal fidelity.: At horizons $k \in \{1, 2, 4, 8\}$ the framework defines grid-based temporal change error and, when correspondence is supplied, motion-compensated $\text{TEPE}_{\text{corr}}$, warped depth DTCE and pure prediction consistency (OPW). Estimated-flow correspondences are marked as flow-conditioned proxies. We do not build on the motion-compensated family here, and say why rather than omit it: it is informative — on held-out D7 both $\text{TEPE}_{\text{corr}}$ and OPW improve in all 20/20 backbone–horizon cases — but on development D2 the two disagree, which needs more space to treat honestly than a submission of this length allows.

c) Intervention harm and benefit.: For $\delta e = e^{\text{fused}} - e^{\text{raw}}$, we measure

$$H^+ = \mathbb{E}[\max(\delta e, 0)], \tag{9}$$

$$B^+ = \mathbb{E}[\max(-\delta e, 0)], \tag{10}$$

$$\text{HUR} = P(\delta e > 0), \tag{11}$$

$$\text{BUR} = P(\delta e < 0). \tag{12}$$

For each bad-pixel threshold τ we additionally report NewBad, RecoveredBad, bad-rate delta, and identity preservation — the fraction of pixels already within τ in the raw prediction that are still within it after refinement. It is a retention rate, not a count of untouched pixels: the convex blend moves almost every pixel on the support, and the question is whether it moves the good ones out.

d) Frame-level tails.: For each frame,

$$D_t = \frac{1}{|M_t|} \sum_{x \in M_t} (e_t^{\text{fused}}(x) - e_t^{\text{raw}}(x)). \quad (13)$$

We retain mean, median, P95, P99, worst degradation, and the fraction of degraded frames.

Where a risk score exists the framework also supports risk–coverage analysis and its calibration statistics [17]. Primary aggregation is equal-sequence macro averaging; pixel-weighted micro statistics are retained secondarily.

V. Results

A. Does the Learned Temporal Policy Matter?

We isolate the learned decision policy on D2. All variants share the frozen stereo predictions, the alignment, the support contract and the evaluation code; only the policy changes, and all run unconfined so the comparison shares one implementation (confinement, Sec. V-J, moves D2 EPE by 1.2×10^{-5} px). The fifteen are the horizon grid $H \in \{1, 2, 4, 6, 8\}$ and continuous recurrence for the learned head; fixed $w \in \{0.1, 0.2, 0.3, 0.5\}$; EMA over two and three frames; forward–backward confidence as the weight; aligned-memory propagation; and warped raw previous at $H{=}1$ — nine of which load no checkpoint. Figures here pool all five backbones and every D2 sequence before taking a ratio, so 0.276202 is one raw mean over fifteen cells; reductions per backbone or per arena average per-cell relative reductions instead.

Raw D2 EPE is 0.276202. Canonical $H = 4$ reaches 0.265028 (4.05%), the best matched fixed blend ($w = 0.5$, algebraically identical to EMA2 here) only 0.274562 (0.59%), and the raw-versus-memory oracle 0.242223 (12.30%): almost seven times the gain of the best fixed blend under an identical state machine. Forward–backward confidence is the informative failure — one of TETHER's own input cues, yet used directly as the fusion weight it worsens EPE by 2.17%. Propagating aligned memory directly also degrades accuracy, so the improvement is not alignment, averaging or smoothing.

B. Why $H = 4$, and Why Recurrence Must Be Bounded

Recurrence must be bounded: with no re-anchor the canonical head loses 62.69% of EPE, and gains stop growing with the horizon, $H = 8$ buying 0.04 points over $H = 4$. $H = 6$ is lower by 0.14%, bought at a cost in intervention risk (B^+/H^+ degrades $2.53 \rightarrow 2.30 \rightarrow 2.04$ while NewBad1 nearly triples), so $H = 4$ is the frozen risk–gain operating point, chosen before D7 was opened. The table also says what the evidence space is for: A2 is behind at every bounded horizon and further behind as it lengthens, yet degrades three times less once the bound is removed.

C. Cross-Backbone Development Behaviour

Canonical $H = 4$ improves all five estimators in mean D2 EPE (Table I), from 2.17% to 8.44%, and all six unseen-backbone sequence groups improve. StereoAnywhere alone does not carry it: excluding it the D2 mean falls to 2.82%, but on D7 the mean excluding it is 4.14%, with RAFT-Stereo alone at 6.10%.

D. Held-Out Cross-Backbone Confirmation

We next evaluate the already frozen $H = 4$ policy on held-out D7. All five backbones improve, including both architectures unseen during temporal-head training: mean backbone EPE decreases $0.4921 \rightarrow 0.4696$, or 4.57%.

The improvement is not restricted to the mean: Bad1 and RMSE decrease for every backbone on both splits, P99 decreases in all ten backbone–split cases and maximum error in nine of ten, most strongly for Stereo Anywhere on D2 (33.27 to 14.88 px). InvalidRate stays zero in all ten SCARED-C backbone–split evaluations — a property of SCARED-C, not of the module: Sec. V-J shows an unmasked output does introduce invalid predictions on external domains, which the support contract of (4) prevents.

The EPE gain is therefore not bought by a worse bad-pixel rate or a broader tail. New bad pixels do appear and are measured below. The joint unseen-backbone / unseen-domain cell is the lower-right block of Table I, read in Section V-H.

E. Seed Variance, and What It Reveals About Selection

We pre-registered a seed study before any result existed: only the seed varies, sequences are the resampling unit, and selecting the best seed is prohibited.

Pooled over five backbones the relative Bad1 reduction is $8.40 \pm 2.53\%$ on D2 and $5.15 \pm 0.35\%$ on held-out D7; EPE gives $3.55 \pm 0.45\%$ and $5.07 \pm 0.44\%$.

The held-out effect is stable: 5.15 ± 0.35 Bad1 and 5.07 ± 0.44 EPE, with paired sequence-level bootstrap intervals $[4.11, 8.02]$ and $[3.42, 5.80]$ and no resample in which refinement failed to help.

The development split is not, and our protocol explains it. Checkpoints are selected on best D2 fused EPE, so on D2 the seeds fall monotonically, $10.78 \rightarrow 8.67 \rightarrow 5.75$, a 5.03 point spread; on D7 the ordering *inverts*, seed 0 becomes the weakest, and the spread collapses to 0.69. Selection optimism sits in the split used to select.

F. Refinement as an Intervention

Across both splits and all five backbones TETHER recovers 2.9–10.4× as many Bad1 pixels as it introduces, with B^+/H^+ between 1.76 and 4.59 and identity preservation above 99.87%: it repairs far more than it breaks. Fig. 2 shows this frame by frame.

G. Head-to-Head Against a Published Stabiliser

BiDAStabilizer [3] is the closest published plug-in stabiliser we could run. We compare on DRENDS because only there is *neither* method in domain: we train on ex-vivo porcine tissue and it has never seen synthetic colonoscopy.

TABLE I

ONE FROZEN TETHER CHECKPOINT, CANONICAL SEED 0, ACROSS FIVE FROZEN ESTIMATORS AND THREE ARENAS: THE DEVELOPMENT SPLIT, THE HELD-OUT SPLIT, AND A DOMAIN THE HEAD NEVER SAW. EVERY CELL IS RAW→TETHER AND LOWER IS BETTER. TWO ESTIMATORS WERE EXCLUDED FROM TEMPORAL-HEAD TRAINING, SO THE LOWER-RIGHT BLOCK IS THE JOINT BACKBONE–DOMAIN SHIFT USUALLY LEFT UNMEASURED. MACRO-SEQUENCE AGGREGATION; SEC. V-E GIVES THE THREE-SEED SPREAD.

		D2 development		D7 held out		DRENDS, out of domain	
Backbone	Status	EPE	Bad1 (%)	EPE	Bad1 (%)	EPE	Bad1 (%)
S^2M^2-S	seen	.2704 → **.2645**	1.39 → **1.29**	.5176 → **.5010**	6.44 → **6.08**	—	—
RAFT-Stereo	seen	.2691 → **.2614**	1.60 → **1.49**	.4671 → **.4397**	6.09 → **5.84**	1.3125 → **1.2414**	58.83 → **55.48**
Stereo Anywhere	seen	.3022 → **.2767**	2.18 → **1.75**	.4774 → **.4449**	6.23 → **5.83**	—	—
CREStereo	unseen	.2668 → **.2578**	1.57 → **1.44**	.5165 → **.4957**	6.35 → **6.11**	1.3027 → **1.2350**	55.17 → **52.51**
Fast-FoundationStereo	unseen	.2725 → **.2647**	1.60 → **1.48**	.4819 → **.4668**	5.92 → **5.68**	1.2838 → **1.2257**	54.88 → **52.25**

TABLE II

HORIZON SWEEP ON D2, EPE REDUCTION (%), POOLED AS ELSEWHERE IN THIS SECTION; HIGHER IS BETTER. A2 IS THE 38-CHANNEL VARIANT OF SEC. V-I.

H	Canonical	A2
1	2.61	2.49
2	3.24	3.02
4	**4.05**	3.69
8	4.09	3.52
continuous	−62.69	**−19.17**

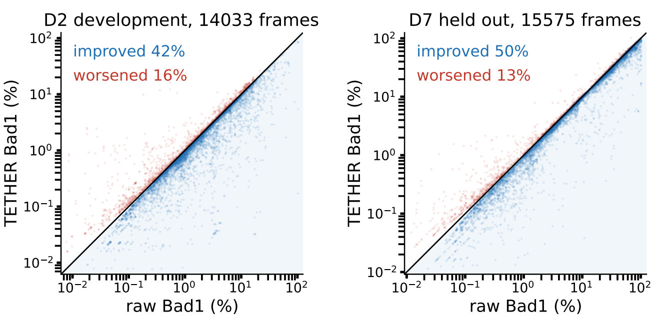

Fig. 2. Per-frame Bad1, raw against refined, for *every* scored frame of each split — five backbones, all sequences, no selection. Points below the diagonal are improved frames. On held-out D7 half improve against 13% worsened, and the worsened band hugs the diagonal while the improved mass extends far from it: the frame-level form of the count-versus-magnitude asymmetry behind B^+/H^+. Disparity *maps* are not shown: the intervention is sub-pixel on disparities of about 8 px, so at an honest colour scale they are indistinguishable.

The shared input is its own RAFT-Stereo *robust* prediction, so TETHER is driven on that exact array over all five recordings (7476 frames) under one support neither method influences. One asymmetry remains and favours the baseline: it is bidirectional and consumes future frames, which our setting forbids.

The regime is a hard one and the Bad1 column says so: DRENDS ground-truth disparity averages 9.89 px and spans 8.42–11.89 between the 5th and 95th percentiles, so a raw EPE of 1.199 px is a third of the scene's entire disparity spread and a 1 px threshold rejects 44.9% of pixels. This is not an artefact of synthetic data: SCARED-C D7 keyframe 3 sits at raw EPE 1.18 px on comparable disparities. What follows compares two refiners of a weak prediction, not a claim that either recovers a good one.

TABLE III

OUT-OF-DOMAIN COMMON SUPPORT ON ALL FIVE DRENDS RECORDINGS, 117.4M PIXELS; *neither* METHOD HAS SEEN THIS DOMAIN. EVERY ROW REFINES THE SAME FROZEN RAFT-STEREO *robust* PREDICTIONS. LOWER IS BETTER.

Method	Causal	EPE	Bad1 (%)	Bad3 (%)	RMSE
Raw	—	1.199	44.92	4.54	1.876
BiDAStabilizer	no	1.175	44.82	4.32	1.776
TETHER	**yes**	**1.133**	**43.52**	**3.98**	**1.699**
reduction vs. raw					
BiDAStabilizer	no	2.0%	0.2%	4.7%	5.3%
TETHER	**yes**	**5.5%**	**3.1%**	**12.3%**	**9.5%**

Out of domain for both, TETHER removes two to thirteen times as much error as the bidirectional baseline on every metric, while being the only causal entry and 54× smaller in trained parameters (177k against 9.64M). It is not uniform: TETHER leads on four recordings of five and on Vid13 is 0.020 px worse. In domain, on three D2 sequences, the same comparison favours us far more (4.58% against 0.25% EPE), which is why we report DRENDS instead.

TC-Stereo [6] needs privileged inputs our interface does not assume: it propagates temporal state by a rigid reprojection from per-frame pose, intrinsics and baseline. SCARED-C *does* supply per-frame poses — correcting them is what the dataset is for — so its temporal path is runnable here. **[TODO: run TC-Stereo's temporal path on the SCARED-C corrected poses; this paragraph claimed the poses did not exist, which was false]** Its frame path alone, moved to the same D2 support, scores EPE 0.524 and Bad1 9.05% against 0.303/2.09% raw and 0.289/1.83% for us.

H. External Zero-Shot Geometry on DRENDS

We evaluate the frozen checkpoint on five DRENDS recordings with three estimators, two excluded from temporal-head training, 20,929 frames. Nothing is adapted, retrained or tuned for this domain.

Across the 15 recording–backbone combinations, disparity EPE, Bad3, metric depth MAE and metric depth RMSE improve in *every* cell (60/60), with mean changes of −5.00%, −17.35%, −3.68% and −3.21%.

The pattern is set by the *recording* and not the estimator: on Vid10 the EPE change is −5.7%, −5.2% and −5.0%

across the seen and the two unseen backbones, and the alignment holds elsewhere, so the residual variance is a property of the data.

Averaged over backbones, the mean EPE reduction is 4.39% for seen and 2.96% for unseen estimators on D2, 4.52% and 3.70% on held-out D7, and 5.31% and 4.84% on DRENDS: every combination is positive, and the shift that costs more is the backbone, not the domain. **[NOTE: pending the S2M2-S and StereoAnywhere DRENDS runs: this seen-backbone mean is n=1, and the paragraph above still says 15 combinations, 60/60 cells, 20,929 frames and three estimators — all five numbers change together]** This establishes zero-shot geometric transfer under a ToF-derived metric reference; it does not establish universal cross-domain robustness, and DRENDS remains a non-independent reference.

I. Ablating Our Own Design Decisions

Four design choices were ablated under a pre-registration naming held-out Bad1 as the endpoint and the three-seed spread as the threshold below which a difference is not read.

TABLE IV

HELD-OUT *relative reduction* (%) AGAINST THE SAME RAW PREDICTIONS, NOT ERROR: HIGHER IS BETTER. CANONICAL AND A2 ARE THREE-SEED MEANS WITH THEIR SPREAD; A1, A3 AND A4 ARE SINGLE RUNS.

Variant	Change	D7 EPE red.	D7 Bad1 red.
Canonical (3 seeds)	—	5.06	5.15 ± 0.34
A1	$142 \rightarrow 78$ ch	4.93	4.88
A2 (3 seeds)	$142 \rightarrow 38$ ch	$\mathbf{5.61 \pm 0.42}$	$\mathbf{5.84 \pm 0.22}$
A3	full-resolution context	5.61	5.85
A4	unconstrained output	0.05	6.02

The result is not the one we expected. A1 is indistinguishable, while A2, A3 and A4 all sit above the canonical mean and A2 and A3 are better on EPE too: none of the four decisions is load-bearing. A2 was then given the canonical model's own three-seed treatment under a rule fixed beforehand, and it stays ahead on both endpoints with a spread eight times tighter (0.32 against 2.53 points of D2 Bad1), so the 142-channel evidence space is not what makes the module work — it is what makes its training unstable. A3 and A4 remain single runs, and A4 disqualifies itself with an EPE gain collapsing to 0.05%.

J. The Support Contract Is Load-Bearing

The results above use the support-confined output of (4); the ablation is a single line, with checkpoint, head, evidence, policy and reset schedule identical. On DRENDS the difference is categorical. Unmasked, depth RMSE *regresses* in $14/14$ cells by $+110.4\%$; confined, it improves in $15/15$ by -3.2%. Disparity follows more quietly, EPE from -4.30% to -5.00%, and invalid refined pixels go from up to 1.3×10^{-5} to exactly zero.

The cause is that zero-padded out-of-bounds memory enters the blend as $\widetilde{m}_t = 0$, the fused disparity collapses there, and bounded recurrence carries it to the next re-anchor. The offenders are rare and bursty — 300 of 38.8M pixels on one recording, in runs of at most H frames — and the aligned memory of every one is exactly 0.0. Since an invalid prediction on valid support is penalised rather than discarded, those few hundred pixels are the entire source of the depth tail previously attributed to the external domain.

On SCARED-C the same ablation is nearly inert: D2 unchanged at 4.05%, D7 from 4.64% to 4.57%. The contract is therefore not free — it costs a little in-domain gain and buys well-posed behaviour out of domain — and it is almost invisible to anyone validating in domain only, which is what makes it worth stating.

a) No-reference temporal behaviour on D4D: We report only prediction-space temporal inconsistency, which TETHER reduces by 53.9% — worth little, since copy-previous removes 97.3% without even aligning and warped-previous scores *exactly* zero by construction. The ordering inverts with respect to geometric quality, so we make no D4D geometric claim.

VI. ANALYSIS

The closure isolates four requirements. Alignment is necessary but not sufficient: the aligned memory carries a 12.30% oracle ceiling, yet using it indiscriminately fails — warped raw memory yields no gain, and both FB-confidence weighting and full recurrent replacement worsen EPE. The mechanism is alignment plus learned graded intervention plus bounded recurrence, not smoothing. The parameter count follows: the head never solves stereo matching, it receives two meaningful candidates and chooses between them.

Ablating that constraint (A4, Table IV) collapses mean EPE gain to 0.05% while improving RMSE and worst case: the failure is broad degradation of pixels already close, not occasional large error. The constraint buys mean accuracy, not safety from large corrections, and we withdraw the attribution of transfer to it.

Two limits are worth stating. TETHER captures 37.2% of the raw-versus-memory oracle gain and 34.5% of the continuous oracle its action space could reach, so the open problem is deciding *where* to intervene. And the effective weight w_t, recovered from the convex form, scores AUROC 0.366–0.382 on D2 but 0.496–0.506 on held-out D7 as a predictor of harmful intervention: an intervention magnitude, not an uncertainty. That does not contradict the intervention ratios above, because the gain factorises exactly into a count and a magnitude term ($B^+/H^+ = 2.839 = 1.642 \times 1.728$ on D7), neither needing w_t to rank anything. The policy is reliably good on average and uninformative per pixel, which makes the oracle gap a ranking problem.

VII. DISCUSSION AND LIMITATIONS

a) What the splits support.: The matched-baseline and horizon closure is development work on D2: it supports mechanism analysis and the pre-frozen $H = 4$ operating point, not confirmation. The frozen checkpoint is evaluated on D7 without changing the policy. Transfer is demonstrated along both axes and their combination, but one external domain does not establish universal robustness, and DRENDS supplies a temporally filtered ToF reference rather than independent stereo ground truth. Per-backbone D7 margins come from one seed and the three-seed study bounds only the pooled effect, so individual rows are not separated from one another.

b) What the metrics cannot say.: No-reference temporal consistency is gameable — replaying warped history scores exactly zero — so we make no D4D geometric claim. Pull-warp support detects out-of-frame sampling but not internal disocclusion, so newly visible anatomy may inherit stale geometry despite plausible motion consistency.

c) Cost and scope.: The head is compact but the deployed system also runs frozen stereo and two SEA-RAFT passes: 33.60 ms overhead at 124.2 MB peak VRAM on an H100. The frozen estimator dominates that budget by up to an order of magnitude, so the end-to-end rate is set by the backbone, from 11.0 FPS on Fast-FoundationStereo (overhead 36.9%) to 1.4 FPS on StereoAnywhere (4.6%). *No real-time claim is made.* We evaluate perception geometry, not downstream control, and demonstrate neither clinical safety nor patient benefit.

VIII. Conclusion

TETHER attaches one causal temporal module, trained once, to frozen and heterogeneous stereo estimators without touching their internals or using future frames. Its gain is not temporal averaging: on development D2 it reduces EPE by 4.05% against 0.59% for the best matched fixed blend under an identical state machine, while direct flow-confidence weighting and unbounded recurrence degrade accuracy, the latter by 62.69%. A frozen $H = 4$ checkpoint improves all five estimators on held-out D7 in EPE, Bad1 and RMSE at once, including two excluded from training, recovers 2.9–$10.4\times$ more Bad1 pixels than it introduces, and holds at $5.15 \pm 0.35\%$ Bad1 across three seeds. Zero-shot transfer to DRENDS improves every one of 60 recording–backbone–metric cells.

Four results bound what may be claimed. The no-reference consistency metric is won outright by a policy that merely replays warped history, so smoothness cannot stand in for correctness. The fusion weight inverts its relation to harm between development and held-out data, so it is an intervention magnitude, not an uncertainty. Development gains carry selection optimism that held-out gains do not: across seeds the D2 effect spans 5.03 points, the D7 effect 0.69. And a module that declares a support must apply it to what it emits: a contract whose violation is almost invisible in domain and costs the whole metric-depth tail out of it, a hazard for flow-aligned plug-in modules generally, not a property of ours.

What remains open is where to intervene: the oracle over the module's own action space is far from reached, and the weight the head computes is a magnitude rather than a ranking.

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